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Evaluation of Mutational Effects on Protein Structures with Alphafold

Introduction

Proteins are involved in many biological processes and depend on the correct three dimensional arrangement of their amino acids. Determining the structure of proteins is a time-consuming and expensive experimental effort. Therefore, in-silico approaches are necessary in order to approximate the effect of potentially significant point mutations on a protein's structure.
The prediction of three dimensional protein structures still remains an ongoing challenge in structural bioinformatics. The critical assessment of techniques for protein structure prediction (CASP) is an international biannual competition aiming to verify the current work on protein structure prediction. During the CASP14 competition in 2020, several models were proposed and tested [1]. Among them Google's DeepMind team introduced their second version of Alphafold [2]. Alphafold outperformed its competitors with a median C_α root mean square deviation (RMSD) of 0.96Å on the backbone structure. We therefore further evaluated the performance of Alphafold for the purpose of mutational effect prediction.

Methods

Alphafold was used to infere several variations of the human hemoglobin beta subunit (Hbb) including the common wildtype as well as the point mutation of sickle cell anemia (p.Glu7Val) [3]. Furthermore, we simulated a set of unknown mutations by systematically replacing a number of consecutive peptides with glutamic. Positions 61 and following were modified to affect the close proximity of the metal binding site at position 64. Glutamic acid was chosen to prevent synonymous substitutions. Additionally, glutamic acid has different chemical properties than most replaced amino acids. Protein structures were compared using the Python module rdkit (Version 2022.3.1). RMSD was calculated on the Cα atoms of the backbone. We compared validated protein structures for the wildtype Hbb and sickle cell anemia - acquired from UniProt[4] - to structures for wildtype and mutated variants of Hbb as predicted by Alphafold.

Results

Table 1 shows the RMSD between various tests performed during the evaluation. Alphafold predicted valid three-dimensional structures for all used peptide sequences. The RMSD of the predicted wildtype Hbb to its reference equals 0.541Å. The predicted sickle cell variant and its reference had an RMSD of 0.911Å. The introduction of the sickle cell mutation (p.Glu7Val) caused a distortion of the Hbb prediction with an RMSD of 0.039Å. Adding artificially constructed mutations increased the RMSD consistently. Three consecutive point mutations resulted in an RMSD of 0.091Å while 15 substitutions resulted in 0.85Å. Figure 1 depicts the predicted 3D structure of the wildtype Hbb and 15 substitutions.

Table 1 RMSD of the C_α atoms of specified proteins. Reference Structures were obtained from UniProt. Substitutions were consecutive replacements of aminoacids with glutamic acid beginning at position 61.

Protein Structure	Protein Structure	RMSD in Å
Reference Wildtype	Reference Sickle Cell	**1.027**
Reference Wildtype	Predicted Wildtype	0.541
Reference Wildtype	Predicted Sickle Cell	0.536
Reference Sickle Cell	Predicted Sickle Cell	0.911
Predicted Wildtype	Predicted Sickle Cell	**0.039**
Predicted Wildtype	1 Substitution	0.060
Predicted Wildtype	3 Substitutions	0.091
Predicted Wildtype	15 Substitutions	**0.850**

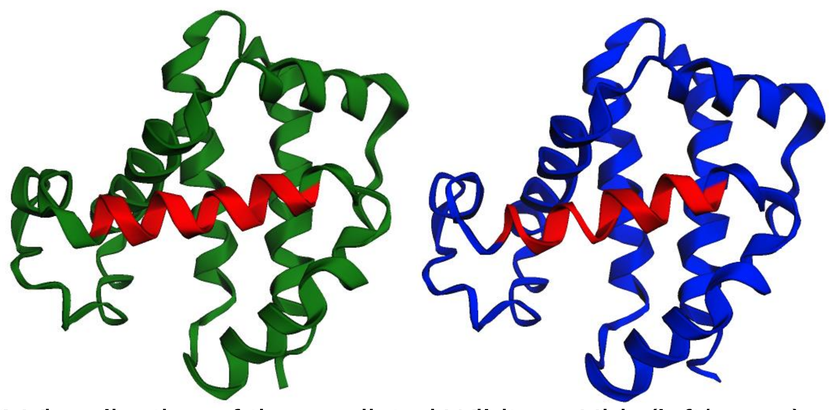

Figure 1 Visualization of the predicted Wildtype Hbb (left/green) and predicted Variant with 15 consecutive Substitutions (right/blue). Alternative Regions are highlighted in red.

Discussion & Conclusion

Alphafold is a robust and accurate model that produces good results on well preserved protein structures. Nevertheless, our results indicate that it is currently not able to cope with point mutations and their global effect on the backbone structure. We propose to adapt Alphafold to increase the impact of point mutations.

Literature/References

[1] Pereira J, Simpkin AJ, Hartmann MD, Rigden DJ, Keegan RM, Lupas AN. Highaccuracy protein structure prediction in CASP14. Proteins. 2021 Dec;89(12):1687-1699. doi: 10.1002/prot.26171. Epub 2021 Jul 14. PMID: 34218458. [2] Jumper, J., Evans, R., Pritzel, A. et al. Highly accurate protein structure prediction with AlphaFold. Nature 596, 583–589 (2021). [3] Herrick, J. B. Peculiar elongated and sickle-shaped red blood corpuscles in a case of severe anemia. Arch. Intern. Med. 6: 517-521, 1910. [4] The UniProt Consortium UniProt: the universal protein knowledgebase in 2021

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